

**EXPERIENTIAL ENGINEERING EDUCATION
PROTOTYPE / DESIGN BUILDING
REPORT
ON**

LPG LEAKAGE DETECTOR

A Report submitted

by

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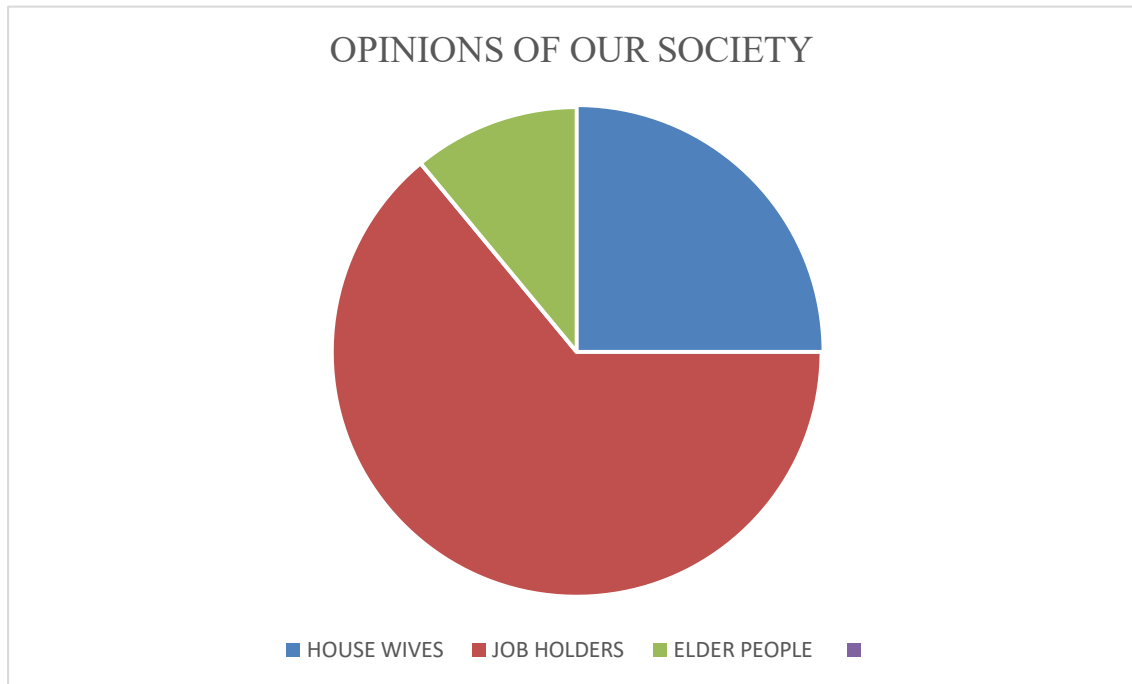


**MECHANICAL ENGINEERING
INSTITUTE OF AERONAUTICAL ENGINEERING
(Autonomous)**

**Dundigal, Hyderabad–500 043, Telangana
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EXEED – PROTOTYPE / DESIGN BUILDING

Define the problem statement and its relevance to today's market / society / industry need.

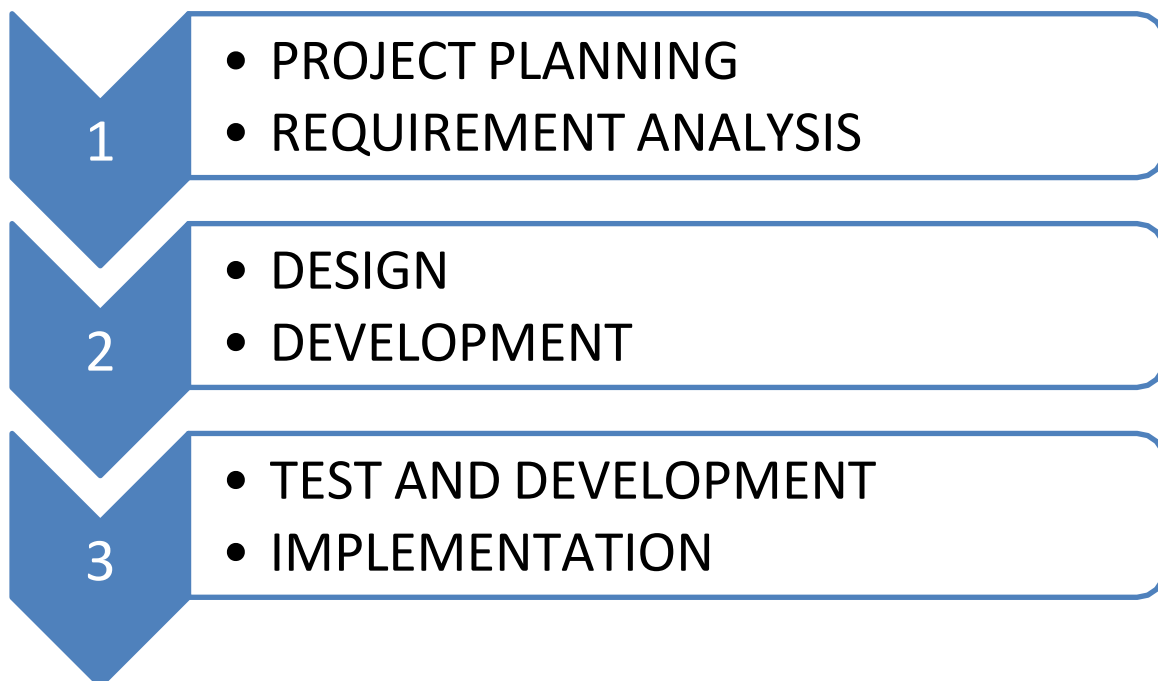


- The leakage of gases only can be detected by human nearby and if there are no human nearby, it cannot be detected. But sometimes it cannot be detected by human that has a low sense of smell. Thus, this system will help to detect the presence of gas leakage.
- LPG Gas leaks have been increased from 0.72% of all kitchen accidents to 10.74% of all the kitchen accidents. The small LPG cylinder of weight 5kg in which the burner is located immediately over the cylinder without using a rubber tube is seen to be safer than the one which uses a rubber pipe as this subway has the hazards of getting cracked which in turn can make way to leakage.

Describe the proposed Solution and Methodology / Developed towards the product/process.

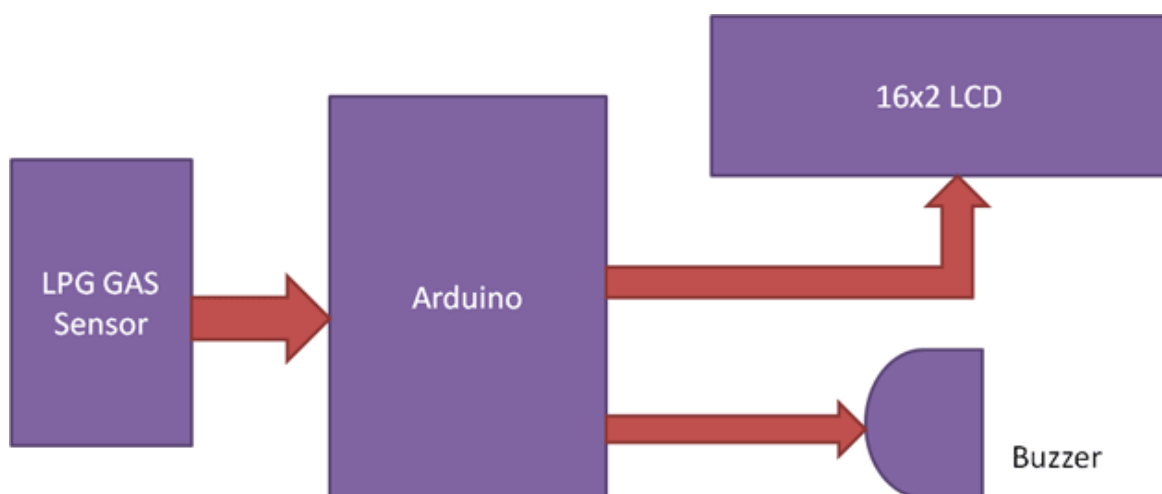
- Methodology is the systematic, theoretical analysis of the methods applied to a field of study. It comprises the theoretical analysis of the body of methods and principles associated with a branch of knowledge. Typically, it encompasses concepts such as paradigm, theoretical model, phases and quantitative or qualitative techniques.
- A methodology does not set out to provide solutions—it is therefore, not the same as a method. Instead, a methodology offers the theoretical underpinning for understanding which method, set of methods, or best practices can be applied to a specific case. Through a methodology, we are achieving the knowledge about planning, design, and implementation and testing.

DESCRIPTION OF METHODOLOGY:



Explain the uniqueness and distinctive features of the product / process / service solution.

- LPG is an essential need of every household, its leakage could lead to a disaster. To alert on LPG leakage and prevent any mishappening there are various products to detect the leakage. Here we have developed an Arduino based LPG gas detector alarm. If gas leakage occurs, this system detects it and makes an alert by buzzing the buzzer attached with the circuit.
- We have used a **LPG gas sensor module** to detect LPG Gas. When LPG gas leakage occurs, it gives a HIGH pulse on its DO pin and arduino continuously reads its DO pin. When Arduino gets a HIGH pulse from LPG Gas module it shows “LPG Gas Leakage Alert” message on 16x2 LCD and activates buzzer which beeps again and again until the gas detector module doesn't sense the gas in environment.

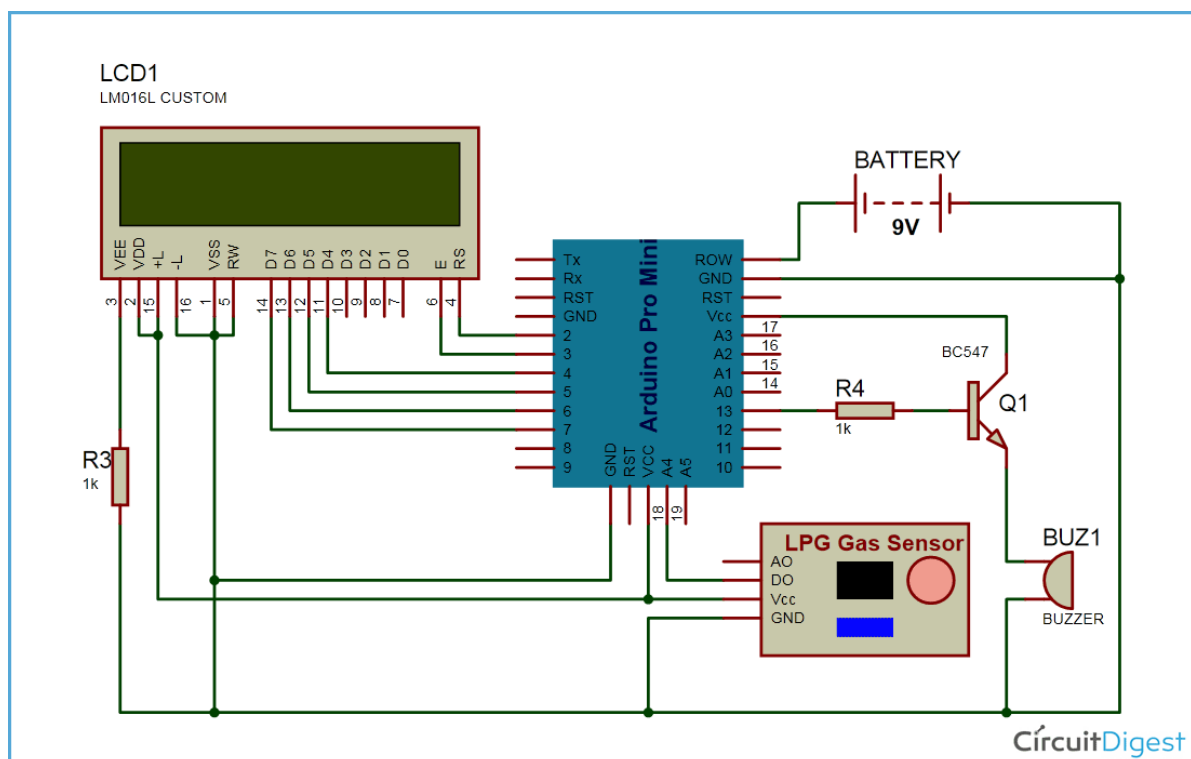


COMPONENTS AND CIRCUIT DIAGRAM OF LPG LEAKAGE DETECTOR.

Components Used:

- Arduino Pro Mini
- LPG Gas sensor Module
- Buzzer
- BC 547 Transistor
- 16x2 LCD
- 1K resistor
- Bread board
- 9-volt battery
- Connecting wires

CIRCUIT DIAGRAM:



Scalability: Highlight the market potential aspects of the Solution/Innovation.

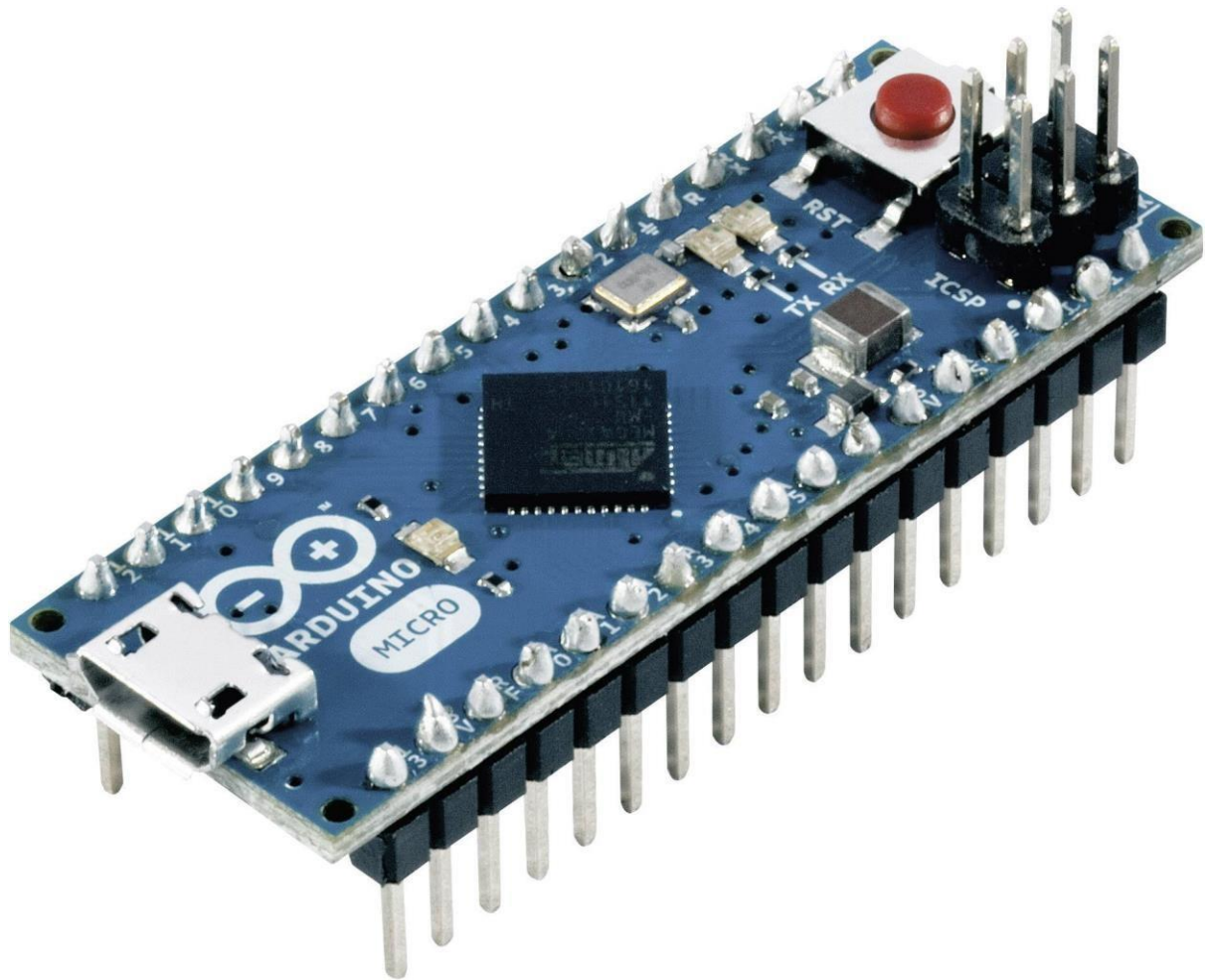
- The gas leak detector market is anticipated to record a CAGR of around 6.5% during the forecast period i.e. 2019-2027. The market is segmented by technology & applications.
- Arduino have the cost of approximately 2.5k and buzzer is approximately cost of RS 273 to 999.
- Our LPG leakage detector costs around 4000rs and this project can help people from LPG leakaging.
- Common precautions include hazard signage, personal protective equipment, and enclosing the testing area so non-essential personnel cannot enter. Proper pipe labeling is essential to ensuring safety during leak detection procedures as well as safety during day-to-day operations.

Details of Prototype:

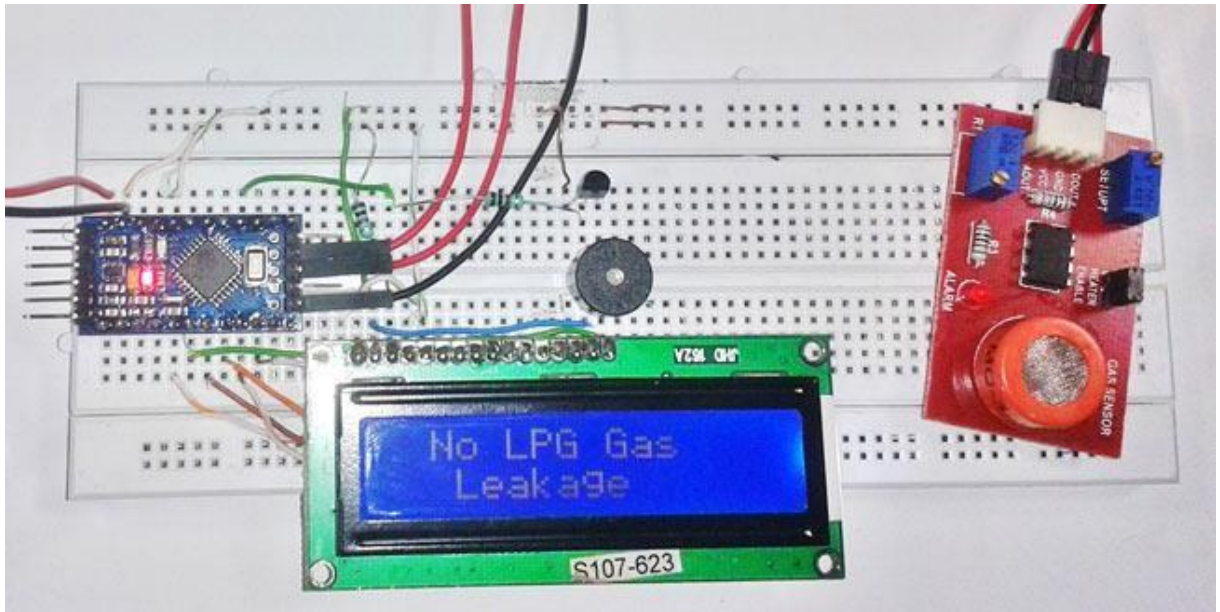
GAS MODULE SENSOR:



ARDUINO:



FINAL PROTOTYPE OF LPG LEAKAGE DETECTOR:



Future Scope in view of the technological development:

- Another major future scope could be including a Automatic Shut-off device which will turn off the gas supply whenever it will detect any gas leakage. This system can be implemented in Industries, Hotels and wherever the LPG cylinders are used.
- In the future days we are planning to modify this project by connecting this project to the main server from which we can send message/notifications to any one in any part of the world using the cloud computing technology and with IOT. Now we can send only mail to the chosen persons.
- When LPG cylinder is empty the cylinder booking will done automatically using IOT technology.

REFERENCES:

- <https://www.ijert.org/research/gas-detection-system-using-arduino-IJERTCONV9IS12062>
- https://www.researchgate.net/publication/347326171_Sensor-Based_Gas_Leakage_Detector_System